

SAFE | SAFE-24

Architectural Runway and Enabler Backlog

The technical foundation that allows future business features to be delivered without delay

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Architectural Runway and Enabler Backlog
Document reference	SAFE-24-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	System Architect
Approver	Enterprise Architect for enterprise-significant enablers
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
SAFE-09	Solution Intent	Records the architecture the runway serves
SAFE-14	Feature Definition	Features depend on runway
SAFE-03	Lean Budgets and Guardrails	Sets enabler capacity allocation

How to use this template

Aspect	Guidance
Purpose	Records the state of the architectural runway — the existing code, components, infrastructure and compliance groundwork that supports near-term features — and the enabler backlog that extends it.
Framework	SAFe — Agile Product Delivery — Architectural Runway; Continuous Learning Culture
Created when	At ART launch; maintained each PI
Reviewed / updated	At every PI planning and at Inspect and Adapt
Owner	System Architect
Approver	Enterprise Architect for enterprise-significant enablers
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	The capacity allocation for enablers is agreed at PI planning with Business Owners. Without a protected allocation, runway erodes and delivery slows a PI or two later — by which point the cause is no longer obvious.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Runway is assessed per upcoming feature, not in the abstract
- Enablers are in the same backlog as features and are WSJF-scored
- Compliance and security enablers are visible, not embedded invisibly in feature work
- Technical debt is quantified in terms of its delivery impact, not just listed
- Capacity allocation for enablers is agreed and tracked against actual

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1. Runway assessment

For each upcoming feature, does the runway exist? Red here means the feature cannot be planned into the next PI.

Feature	Target PI	Runway required	Exists?	Gap	Enabler ref
FE-105 admission control, and every subsequent platform release	PI 4	No signing identity or admission policy exists on the pilot cluster	In progress	No signing identity or admission policy exists on the pilot cluster	Signed container image supply chain
Trusted expansion of the estate in Tranche 3	PI 5	Vendor attestation API not yet generally available	Not started	Vendor attestation API not yet generally available	Firmware attestation at node admission
Per-tenant credential isolation without static secrets	PI 5	Dependent on the Enterprise Identity Programme (DEP-003)	Blocked	Dependent on the Enterprise Identity Programme (DEP-003)	Workload identity federation
FE-109 GPU telemetry and the utilisation benefit measure	PI 4	Closed — capacity uplifted in PI 3	Done	Closed — capacity uplifted in PI 3	Observability ingest capacity uplift
FE-102 pre-emption without loss of work	PI 4	Only framework-native checkpointing works today	In progress	Only framework-native checkpointing works today	Checkpoint-and-resume for distributed jobs
FE-113 and the four-hour recovery objective	PI 6	Control plane is currently single-hall	Not started	Control plane is currently single-hall	Cross-hall control plane for failover
FE-110 lineage and experiment reproducibility	PI 5	Datasets are mutable on the current storage tier	Not started	Datasets are mutable on the current storage tier	Immutable dataset versioning
Continuous assurance rather than point-in-time review	PI 6	Evidence assembled manually for each assurance milestone	Not started	Evidence assembled manually for each assurance milestone	Automated compliance evidence generation

2. Enabler backlog

Ref	Enabler	Type	Enables	WSJF	Target PI	Owner	Status
Funded from the Tranche 2	Signed container image supply chain	Architectural / Infrastructure / Compliance / Exploration	FE-105 admission control, and every subsequent platform release	14.2	PI 4	K. Adeyemi, Security Architect	In progress

Ref	Enabler	Type	Enables	WSJF	Target PI	Owner	Status
allocation of AED 118,000,000							
EB-02	Firmware attestation at node admission	Architectural	Trusted expansion of the estate in Tranche 3	9.4	PI 5	K. Adeyemi, Security Architect	Not started
EB-03	Workload identity federation	Architectural	Per-tenant credential isolation without static secrets	11.1	PI 5	K. Adeyemi, Security Architect	Blocked
EB-04	Observability ingest capacity uplift	Infrastructure	FE-109 GPU telemetry and the utilisation benefit measure	8.7	PI 4	G. Lindqvist, Service Operations Lead	Done
EB-05	Checkpoint-and-resume for distributed jobs	Exploration	FE-102 pre-emption without loss of work	7.9	PI 4	P. Raghavan, Platform Engineering Lead	In progress
EB-06	Cross-hall control plane for failover	Architectural	FE-113 and the four-hour recovery objective	6.4	PI 6	T. Nakamura, Chief Platform Architect	Not started
EB-07	Immutable dataset versioning	Architectural	FE-110 lineage and experiment reproducibility	6.1	PI 5	I. Petrova, Data Platform Lead	Not started
EB-08	Automated compliance evidence generation	Compliance	Continuous assurance rather than point-in-time review	5.3	PI 6	L. Haddad, CISO	Not started
EB-09	Signed container image supply chain	Compliance	FE-105 admission control, and every subsequent platform release	14.2	PI 4	K. Adeyemi, Security	In progress

Ref	Enabler	Type	Enables	WSJF	Target PI	Owner	Status
						Architect	

3. Technical debt register

Quantify the cost. "It slows us down" is not a case for investment; "it adds two days to every release" is.

Ref	Debt item	Impact on delivery	Cost to fix	Cost of not fixing	Priority
TDR-01	400G fabric commissioned and NCCL-validated	Contingency drawdown of 12% against the AED 9.4m held	AED 1,120,000	AED 240,000	High
TDR-02	Platform v1.0 — scheduling, quota, observability and image supply chain	Training throughput approximately 9% below the business case assumption	AED 91,200,000	AED 17,800,000	Medium
TDR-03	Hall A accepted at 1,024 GPUs with acceptance certificate	Reportable data incident and loss of regulatory standing	AED 4,200,000	AED 1,120,000	Must have
TDR-04	Independent security assurance report with findings closed	Critical path delay of up to ten weeks and a quarter of deferred benefit	AED 96,000	AED 91,200,000	Should have

4. Non-functional requirements backlog

NFRs that require runway work rather than feature work.

Ref	NFR	Current state	Required state	Enabler ref
NFR-01	All container images signed and admitted only from the internal registry	Met	Not started	Workload identity federation
NFR-02	No standing vendor privilege; all access brokered and recorded	Partially met	Complete	Observability capacity uplift
NFR-03	Restricted data remains within UAE borders at rest and in transit	Met	In progress	Firmware attestation at node admission
NFR-04	All-reduce efficiency at or above 92% at 256-GPU scale	Not met — 84%	Open	Signed image supply chain
NFR-05	Storage sustains 1.2 TB/s aggregate read	Met	Not started	Cross-hall failover control plane
NFR-	Scheduled workloads	Not yet tested	Complete	Workload identity

Ref	NFR	Current state	Required state	Enabler ref
06	recover within four hours of a single hall failure			federation

5. Capacity allocation tracking

PI	Enabler allocation agreed %	Actual %	Variance	Consequence
Constrained by the 6.2 MVA site limit at DXB-1 until 2028	Yes	31%	+AED 1,120,000	Hall A capacity capped at 1,536 GPUs until resolved
Subject to closure of ISS-003 before multi-tenant go-live	Partially	48%	-AED 340,000	Critical path delay of up to ten weeks and a quarter of deferred benefit
Reported through the monthly benefit tracking report	Yes	72%	Nil	Reportable data incident and loss of regulatory standing
Applies from hall A handover; hall B follows in Tranche 2 closure	No	92%	+2 weeks	Training throughput approximately 9% below the business case assumption

6. Architectural decisions this PI

Summary only — the detail belongs in solution intent.

ID	Decision	Date	Impact on runway
DEC-001	<i>Build a shared central AI compute platform rather than continue funding departmental clusters</i>	12 Mar 2026	<i>On-call rota cannot be sustained beyond a 5x8 model at handover</i>
DEC-002	<i>Adopt liquid cooling with rear-door heat exchangers rather than air cooling at reduced density</i>	28 Apr 2026	<i>Contingency drawdown of 12% against the AED 9.4m held</i>
DEC-003	<i>Single scheduling plane — Slurm workloads run under the Kubernetes scheduler via a bridging operator</i>	19 May 2026	<i>Training throughput approximately 9% below the business case assumption</i>
DEC-004	<i>DXB-1 is the primary site; AUH-2 is deferred to Tranche 3</i>	30 Jun 2026	<i>Contingency drawdown of 12% against the AED 9.4m held</i>
DEC-005	<i>All container images must be signed and admitted only from the internal registry</i>	11 Aug 2026	<i>On-call rota cannot be sustained beyond a 5x8 model at handover</i>